SHAHAZAD ABDULLA

Kottayam, Kerala, India

+91 7736652034 | shahazad.abdulla.engineer@gmail.com | linkedin.com/in/shahazad-abdulla | github.com/ShahazadAbdulla

Summary

Results-driven Robotics and Automation Engineer with extensive R&D experience at **CSIR-NAL** and **Intel**. Proven expertise in designing and implementing robust autonomous systems, specializing in **safety-critical, end-to-end systems integration**, from low-level IoT data pipelines to high-level perception including VSLAM and sensor fusion. Recognized as a **Top 3 global contributor** to a major ROS 2 release and a key developer on the **Team Equinox (BAJA SAEINDIA)** autonomous vehicle team. Seeking a challenging internship to apply advanced skills in full-stack robotics systems integration.

Education

Saintgits College of Engineering (Autonomous) — Kottayam, Kerala, India
Bachelor of Technology in Robotics and Automation Engineering — Expected July 2027

Skills

- **Programming:** Python, C++, C, MATLAB, Bash
- **Robotics Software:** ROS 2, Gazebo, Isaac Sim, Navigation2, SLAM (RTAB-Map, VSLAM), Sensor Fusion (EKF), Digital Twins (PyBullet), micro-ROS, Control Systems (PID), URDF/SDF/XACRO
- **Hardware & Comms:** Hardware: NVIDIA Jetson, Microcontrollers (Arduino, ESP32), Motor Controllers, DACs, IMUs, RADAR
  Comms & Protocols: CAN bus, MQTT, UDP, I2C, Serial
- **Computer Vision & ML:** OpenCV, PyTorch, TensorFlow/Keras, YOLO (v5/v8), OCR (EasyOCR), Data Augmentation, Camera Calibration, cv_bridge
- **Tools & Platforms:** Linux, Git, GitHub, Docker, CMake, colcon, ESP-IDF, LaTeX

Professional & Technical Experience

CSIR - National Aerospace Laboratories (NAL) — Bengaluru, India
Research Intern, Project Orion — June 2025 – July 2025

- Architected and validated a multi-sensor EKF-fused SLAM system for an autonomous inspection UGV, **reducing end-of-path trajectory error by over 90%** compared to a VIO-only baseline.
- Engineered a complete perception and control stack on an NVIDIA Jetson Orin Nano, from low-level C++ driver modification to high-level SLAM implementation in ROS 2.
- Solved critical real-world integration challenges, including numerical instability of the IMU and kinematic mismatches, by developing a novel data-driven calibration methodology (SBKC).
- Co-authored the research paper, "AUTOMATING THE VISUAL INSPECTION OF AIRCRAFT STRUCTURES USING 2D AND 3D COMPUTER VISION, MACHINE LEARNING, AND ROBOTICS." published in the *International Conference on Advances in Robotics, Machines and Structures (ARMS 2025).*

Intel Corporation (Intel Unnati Program) — Kottayam, Kerala, India
Industrial Trainee - Quality Control Automation — May 2025 – July 2025

- Architected a full-stack, ROS 2-based system for AI-driven validation, featuring a PyBullet digital twin, a responsive UI for monitoring, and an SQLite backend for robust data logging.
- Designed and implemented a multi-stage AI pipeline featuring OCR (EasyOCR), QR code decoding, and a custom-trained TensorFlow/Keras model for print quality assessment.
- Achieved **100% precision on "Bad Label" classification** by tuning the ML model's operating threshold, guaranteeing no defective products were passed by the quality check.

Team Equinox (BAJA SAEINDIA) Kottayam, Kerala, India
Vice Captain (2026 Season) Nov 2025 – Present
Lead Autonomous Systems Integration Engineer (2025 Season) Mar 2024 – Oct 2025

- Spearheaded the autonomous systems integration, culminating in an **All India Rank of 7th** at the aBAJA 2025 competition and flawless passes on all technical and autonomous checks on the first attempt.
- Architected the vehicle's distributed, real-time control architecture; integrated a high-level NVIDIA Jetson with multiple low-level Arduino R4 Minima nodes over a robust **CAN bus** network.
- Engineered safety-critical firmware, including PID controllers for the throttle, brake, and steering actuators, ensuring precise vehicle control and adherence to competition rules.

Taiwan-India Smart Cities Project (Saintgits & CCU, Taiwan) Kottayam, India
Lead IoT Systems Engineer Aug 2025 – Present

- Coordinated the development and deployment of prototype IoT sensor nodes for the world's first cross-nation federated AI learning platform.
- Engineered the embedded software for ESP32 microcontrollers to reliably publish real-time environmental data via MQTT to the central data server.
- Led a student team in the hands-on building, testing, and debugging of the hardware nodes, ensuring data integrity for the AI backend.

Open Source Contributions

The ROS 2 Community Remote
ROS 2 Contributor May 2024 – Present

- Ranked **3rd globally** in the ROS 2 Kilted Kaiju "Tutorial Party" by validating and closing **over 51 critical issues**, receiving official acknowledgment from Open Robotics.
- Actively contribute to the `ros2_control` framework, addressing issues and improving the core control architecture for the ROS 2 community.

MPU6050 Library for Aries Board Open Source Project
Creator & Maintainer Jan 2025 – Mar 2025

- Engineered and open-sourced a custom MPU6050 sensor library in C++, acknowledged by Vega Processors for a significant contribution to their RISC-V hardware ecosystem.

Selected Technical Projects

- **Autonomous Lane-Keeping Perception System:** Developed a ROS 2 perception node by integrating the Ultrafast Lane Detection (UFLD) model to accurately identify lane boundaries from real-world video feeds.
- **End-to-End Object Detection Pipeline in Simulation:** Architected and implemented a complete ROS 2 perception pipeline in Gazebo, successfully streaming data from a simulated TurtleBot3 camera to a YOLOv5 node and visualizing real-time object detections in RViz2.
- **Self-Balancing Robot with Kalman Filter:** Engineered a complete control system in C++ for a physical self-balancing robot on a RISC-V board, implementing a Kalman filter and PID controller for stable state estimation.

Leadership & Achievements

- **BAJA SAEINDIA National Competition (2025):** Secured **All India Rank 7** as the Lead Autonomous Systems Integration Engineer for Team Equinox.
- **Aspire Leaders Program (2025):** Selected from over 26,000 global applicants for a competitive leadership program developed by the Aspire Institute with Harvard Business School faculty.
- **University Leadership:** Vice Chairman, The Robotics Society Saintgits Chapter; Co-Lead, TinkerHub Saintgits.